YEAR 9 MATHS · CHAPTER 10 · MIND MAP

Quadratic equations and graphs

tuterly.com.au · Free for students and families

1 Quadratic equations

Null Factor Law: if $ab = 0$, then $a = 0$ or $b = 0$.

$(x - 3)(x + 5) = 0$

$x = 3$ or $x = -5$

A quadratic can have two, one or no solutions.

2 Solving $ax^2 + bx = 0$ and $x^2 = d$

Take out the common factor. For $x^2 = d$, remember $\pm$.

$2x^2 - 6x = 0 \Rightarrow 2x(x - 3) = 0 \Rightarrow x = 0$ or 3

$x^2 = 16 \Rightarrow x = \pm 4$

Watch out: don't divide by x: you'd lose $x = 0$.

3 Solving $x^2 + bx + c = 0$

Make one side 0, factorise, then use the Null Factor Law.

$x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0$, so $x = -4$ or 3

$x^2 = 5x - 6 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2$ or 3

4 Using quadratics to solve problems

Write an equation, solve it, then reject answers that don't make sense.

A rectangle is 3 m longer than it is wide, area 40 m²: $x(x + 3) = 40$

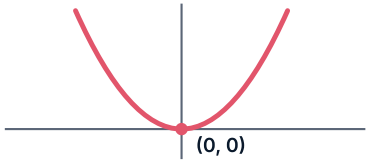
$x^2 + 3x - 40 = 0 \Rightarrow (x + 8)(x - 5) = 0$

Width can't be -8, so it's 5 m by 8 m

5 The parabola

$y = x^2$ is a parabola. It has a turning point and an axis of symmetry.

$a > 0$: opens up (minimum). $a < 0$: opens down (maximum).

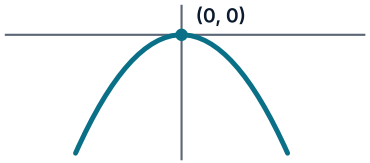


6 Dilations and reflections

$y = ax^2$: a bigger $|a|$ makes it narrower; a negative a flips it upside down.

$y = 3x^2$: narrower $y = \frac{1}{2}x^2$: wider

$y = -x^2$: reflected in the x-axis



7 Translations

$y = (x - h)^2 + k$ has turning point (h, k).

$y = (x - 2)^2 + 3$: turning point $(2, 3)$

$y = (x + 1)^2 - 4$: turning point $(-1, -4)$

Watch out: the sign of h flips: $(x + 1)$ moves left 1.

8 Intercept form

$y = (x - a)(x - b)$ crosses the x-axis at a and b. The turning point is halfway between.

$y = (x - 1)(x - 5)$: x-intercepts 1 and 5, axis $x = 3$

Turning point $y = (3 - 1)(3 - 5) = -4$, so $(3, -4)$; y-intercept $(0, 5)$

